



Unveiling the orientation sensitivity of creep life in near [001] oriented Ni-based single crystal superalloys at intermediate temperatures

Pengfei Qu, Wenchao Yang^{*}, Qiang Wang, Chen Liu, Jiarun Qin, Jun Zhang, Lin Liu

State Key Laboratory of Solidification Processing, Northwestern Polytechnical University, Xi'an 710072, China

ARTICLE INFO

Keywords:

Ni-based single crystal superalloys
Creep
Orientation sensitivity
Orientation rotation

ABSTRACT

Although blades with a deviation angle of less than 15° between the blade stacking axis and the [001] orientation are qualified in the industry, the creep life of samples near [001] orientation exhibits significant anisotropy at intermediate temperatures. Those crystals having orientations within 15° from precise [001] exhibited significantly longer lives when their orientations were closer to the [001]-[101] boundary of the stereographic triangle than to the [001]-[111] boundary. Here, we first investigated the orientation rotation path of specimens near [001] orientation during creep at $750^\circ\text{C}/750\text{ MPa}$, then revealed the dominant slip systems at different creep stages. Subsequently, we evaluated the effect of orientation deviation from precise [001] on creep properties. Finally, our research revealed the orientation sensitive mechanism of creep life in near [001] oriented Ni-based single crystal superalloys at intermediates.

1. Introduction

To increase the thrust-to-weight ratio and efficiency of aircraft engines, the turbine entry temperature continuously increases, resulting in an increasingly harsh service environment for turbine blades. Then the harsh environment would result in turbine blades operating under extreme cyclic conditions over a range of temperatures and complex stress states, in which the working temperature of the blade root, tenon, and inner wall of the air-cooled channel is intermediate temperatures ($650^\circ\text{C}\sim 850^\circ\text{C}$) (Darolia, 2019; Dye et al., 2008; Kim et al., 2016). Therefore, the creep performance at intermediate temperatures is one of the key factors in the material selection of turbine blades.

Ni-based single crystal superalloys are considered the preferred material for turbine blades due to their excellent comprehensive properties (Mukherjee et al., 2022; Reed, 2006). Considering [001] crystal orientation is the preferred direction of crystal growth in Ni-based superalloys and its great comprehensive performance, single crystal turbine blades usually are directionally solidified during the casting process with the crystallographic orientation [001] aligned with the blade stacking axis (Tan et al., 2018; Zhou et al., 2019). However, during the directional solidification, the blade stacking axis does not always yield precisely the [001] orientation (Tan et al., 2018; Xiao et al., 2022). Fig. 1a describes the relationship between the single crystal blade coordinate system and the three-dimensional orientation of Ni-based single crystal superalloys, in which the uppercase variables X, Y, and Z represent the blade

^{*} Corresponding author.

E-mail address: wenchaoyang@nwpu.edu.cn (W. Yang).

coordinate system and the lowercase variables x , y , and z represent the three-dimensional orientation of superalloys. There is a deviation angle (θ) between the blade stacking axis (Z) and the $[001]$ orientation (z), and deviations up to 15° are commonly accepted within the industry (Wang et al., 2023; Yu et al., 2020). Therefore, it is necessary to study the creep properties of single crystals deviating from the $[001]$ orientation within 15° at intermediate temperatures.

The creep behavior of Ni-based single crystal superalloys exhibits significant anisotropy at intermediate temperature, and the creep life of samples near $[001]$ orientation are particularly sensitive to crystal orientation (Heep et al., 2022; Li et al., 2022c; MacLachlan et al., 2001; Qu et al., 2024a; Yu et al., 2020, 2013). Mackay et al. (MacKay and Maier, 1982; Mackay et al., 1980) systematically investigated the creep rupture properties of MAR-M247 with various crystal orientations at $774^\circ\text{C}/724\text{ MPa}$, and established the relationship between orientation and creep rupture life at intermediate temperature. As shown in Fig. 1b, in the vicinity of $[001]$ orientation, the creep properties exhibit more obvious orientation sensitivity, and the creep life of samples near the $[001]$ - $[101]$ boundary was much longer than those close to the $[001]$ - $[111]$ boundary. The creep life of other superalloys with near $[001]$ orientation also follows the pattern shown in Fig. 1b (Heep et al., 2022; Li et al., 2022c; Sass et al., 1996; Yu et al., 2020). Therefore, the deviation angle of 15° from precise $[001]$ is not an appropriate indicator to determine whether the blade is qualified, and a deviation angle of 15° from $[001]$ in $[101]$ directions can be tolerated, while a deviation towards the $[111]$ direction cannot be accepted. Many researchers have been dedicated to revealing the intrinsic mechanism of orientation sensitivity for the creep properties of near $[001]$ -orientated superalloys at intermediate temperatures. Leverant and Kear (1970) first found the difference in creep rupture behavior among crystals having orientations along the $[001]$ - $[111]$ and $[001]$ - $[101]$ boundaries in the MAR-M200 single crystal at 760°C and believed that it was correlated with the Schmid factors and multiplicity of slip for the $\{111\}$ $\langle 112 \rangle$ slip systems. Subsequently, Mackay et al. (MacKay and Maier, 1982; Mackay et al., 1980) proposed the creep anisotropy was largely influenced by the orientation rotation in the primary stage of creep, and they also believed that the orientation rotation was caused by the activity of $\{111\}$ $\langle 112 \rangle$ slip system. Although more results also suggested the activity of a single $\{111\}$ $\langle 112 \rangle$ slip system would lead to orientation rotation, the rotation did not only occur in the primary stage (Qu et al., 2023a, 2024a; Shi et al., 2022). Regarding the active slip systems, Ghosh et al. (1990) had different views, suggesting that the creep anisotropy was related to $\{111\}$ $\langle 110 \rangle$ and $\{001\}$ $\langle 110 \rangle$ slip systems rather than $\{111\}$ $\langle 112 \rangle$ slip systems. However the slip systems they thought could be active were inconsistent with deformation mechanisms dominated by stacking faults. Sass et al. (1996) thought that the poor creep life of samples near the $[001]$ - $[111]$ boundary was related to extensive single slip on $\{111\}$ $\langle 112 \rangle$ slip systems or coplanar slip on $\{111\}$ $\langle 110 \rangle$ slip systems resulting in high primary creep rates and creep strains. Yu et al. (2020) also thought the mechanism of anisotropic creep was attributed to the heterogeneous γ' precipitate deformation by $\{111\}$ $\langle 112 \rangle$ slip. However, they failed to notice the influence of orientation rotation. In addition, some researchers were more concerned about the influence of $\{111\}$ $\langle 110 \rangle$ slip systems (Heep et al., 2022; Kakehi et al., 1993). Kakehi et al. (1993) also had similar findings and ascribed the effect of crystallographic loading directions on creep rates to different interface dislocation network configurations. Recently, Heep et al. (2022) proposed a new perspective that the creep anisotropy did not rely on dislocation networks or multiple planar fault slips, but was governed by dislocation reactions between γ -channel dislocation families with different Burgers vectors ($a/2\langle 110 \rangle$) on the early stages creep strain accumulation. Unfortunately, they only considered the influence of a single type of slip system, $\{111\}$ $\langle 112 \rangle$ or $\{111\}$ $\langle 110 \rangle$. In addition, they have mostly focused on creep fracture or the primary stage of creep, lacking analysis of different stages of creep. In fact, the activity of the $\{111\}$ $\langle 112 \rangle$ slip system is closely related to the $\{111\}$ $\langle 110 \rangle$ slip system (Drew et al., 2004; Qu et al., 2021; Rae and Reed, 2007; Zhang et al., 2024b). Therefore, when studying the orientation sensitivity for the creep properties in near $[001]$ oriented Ni-based single crystal superalloys, it is necessary to

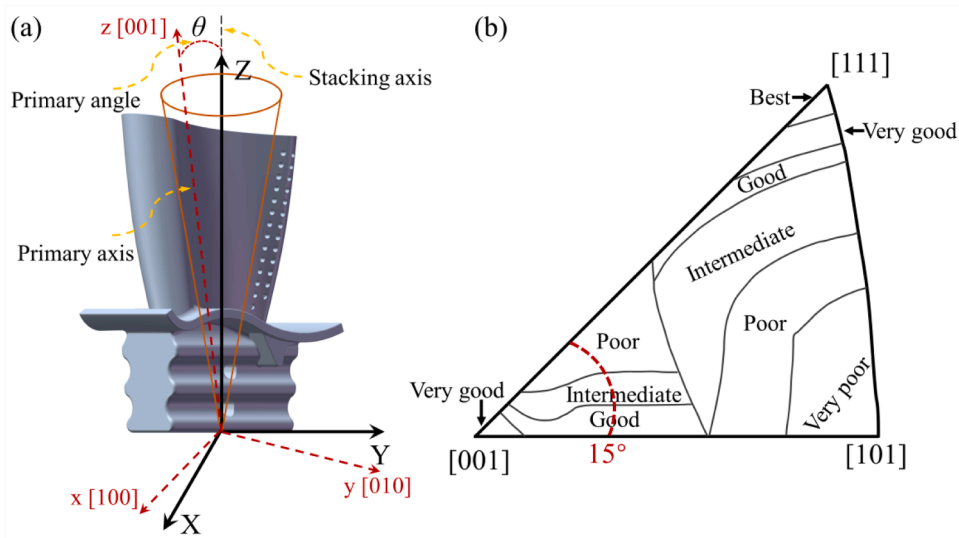


Fig. 1. (a) Relationship diagram between single crystal blade coordinate system and three-dimensional orientation of single crystal superalloys; (b) Relationship between orientation and creep rupture life for single crystal superalloys at intermediate temperature.

comprehensively consider the effect of different slip systems, and distinguish the stages of action between the two types of slip systems.

In this work, a series of creep interruption experiments were conducted on nearly [001] oriented single crystals at 750 °C/750 MPa. A key analysis was conducted on the relationship between microstructure evolution, slip system activation, and orientation rotation at different creep stages. Finally, we revealed the intrinsic mechanism of the orientation sensitivity for creep life in near [001] oriented Ni-based single crystal superalloys at intermediate temperatures.

2. Experimental

A third-generation Ni-based single crystal superalloy is used in the present experiment (Qu et al., 2023b), and the nominal composition of the experimental Ni-based single crystal superalloy is as follows (wt.%): 3.5Cr, 12Co, 1.5Mo, 6 W, 8Ta, 4Re, 6Al, and the balance Ni. The actual composition is as follows (wt.%): 3.46Cr, 12.08Co, 1.47Mo, 6.03 W, 8.02Ta, 3.96Re, 5.98Al, and the balance Ni. Single crystal rods were produced by using a modified Bridgman casting furnace. A stepwise solution heat treatment at 1290 °C/ 1 h + 1300 °C/ 1 h + 1310 °C/ 3 h + 1320 °C/ 5 h + 1328 °C/ 15 h, and then air cooling (A.C.), which was to diminish inter-dendritic eutectic microstructures and solidification segregation (Fuchs, 2001; Zhang et al., 2017). Subsequently, a two-stage ageing treatment was adopted and it was 1180 °C/ 4 h (A.C.) + 870 °C/ 24 h (A.C.). The primary ageing process is to obtain uniformly dispersed γ' phases, and the secondary ageing could adjust the cubic degree of γ' phases. After the standard heat treatment, the sample rods were processed into creep samples. Then, the uniaxial constant load creep interruption testing was carried out in a CTM105-A1 creep-testing machine at 750 °C/750 MPa. First, a complete creep curve was obtained through tensile creep fracture testing. Then, based on the creep curve, the number of samples and interruption time that need to be tested for creep interruption was determined. During creep testing, the heating rate is 10 °C/min and the insulation time is 1 h. Samples would be cooled by air cooling after creep fracture or interruption. Samples, which were all perpendicular to the tensile axis, were cut for orientation analysis. A Tescan Clara GMH, equipped with an electron back-scattered diffraction (EBSD) detector (Oxford Symmetry Model), was used for EBSD test. EBSD test was conducted at an accelerating voltage of 20 kV, current of 10 nA, and a step size of 1 μ m. After EBSD testing, AZtecCrystal software was used for subsequent data analysis. Finally, an FEI Talos F200X TEM, operating at 200 kV, was used for TEM investigations. Samples, for

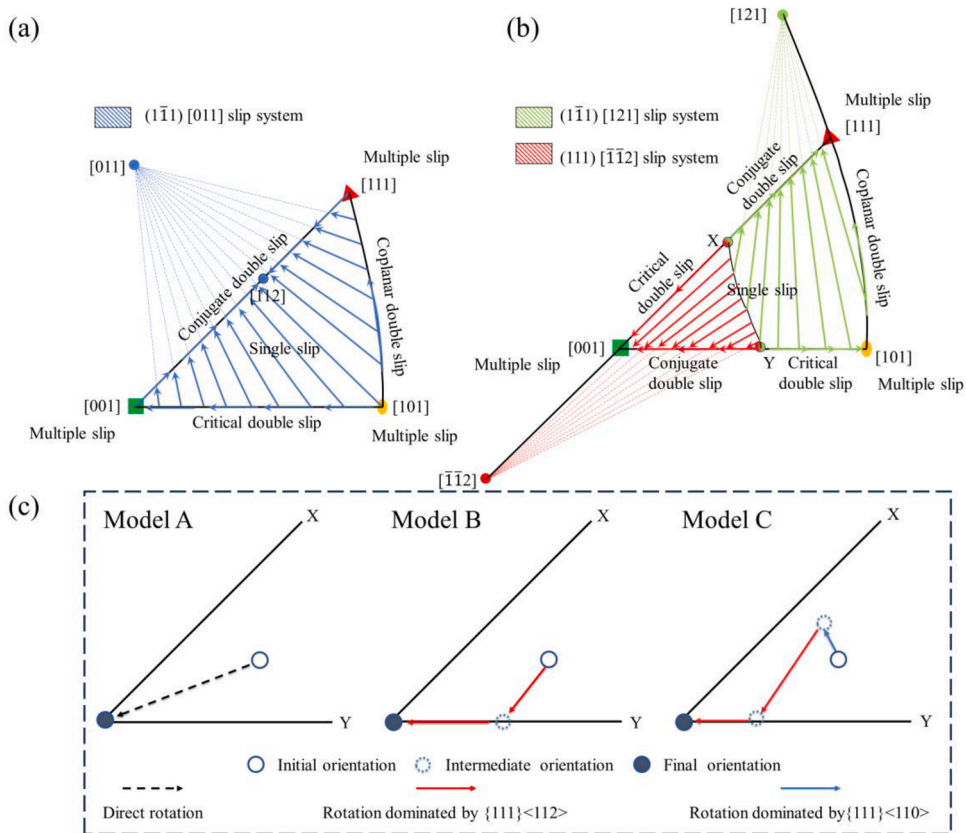


Fig. 2. (a) Illustration of lattice rotation process dominated by $\{111\}$ $\langle 110 \rangle$ slip systems; (b) Illustration of lattice rotation process dominated by $\{111\}$ $\langle 112 \rangle$ slip systems; (c) Possible orientation rotation modes of near [001] oriented samples during tensile creep process: Model A is the direct rotation model; Model B is the model that the rotation is dominated by $\{111\}$ $\langle 112 \rangle$ slip system; Model C is the model that the rotation is dominated by both $\{111\}$ $\langle 110 \rangle$ and $\{111\}$ $\langle 112 \rangle$ slip systems.

TEM observation, were mechanically ground to 60 μm firstly and then were electrochemically thinned using a twin-jet polisher with an electrolyte of 7 % perchloric acid in ethanol at $-25\text{ }^{\circ}\text{C}$, 25 mA.

3. Feasibility analysis of different orientation rotation models

Although various theories have been proposed to explain the orientation sensitivity of creep life near [001]-orientated single crystal superalloys at intermediate temperatures, the orientation rotation theory proposed by MacKay and Maier (1982) seems to be the most reasonable. However, the effect of $\{111\}\langle 110\rangle$ slip systems on orientation rotation has been ignored. In fact, the activity of the $\{111\}\langle 112\rangle$ requires (i) sufficient densities of at least two $a/2\{111\}\langle 110\rangle$ families of dislocations in γ matrix, to enable the nucleation of the dislocation ribbons in accordance with the reaction given above, and (ii) sufficient resolved shear stress for the propagation of the $a\langle 112\rangle$ ribbon (Qu et al., 2021; Rae et al., 2001, 2000). Therefore, the orientation rotation should comprehensively consider the

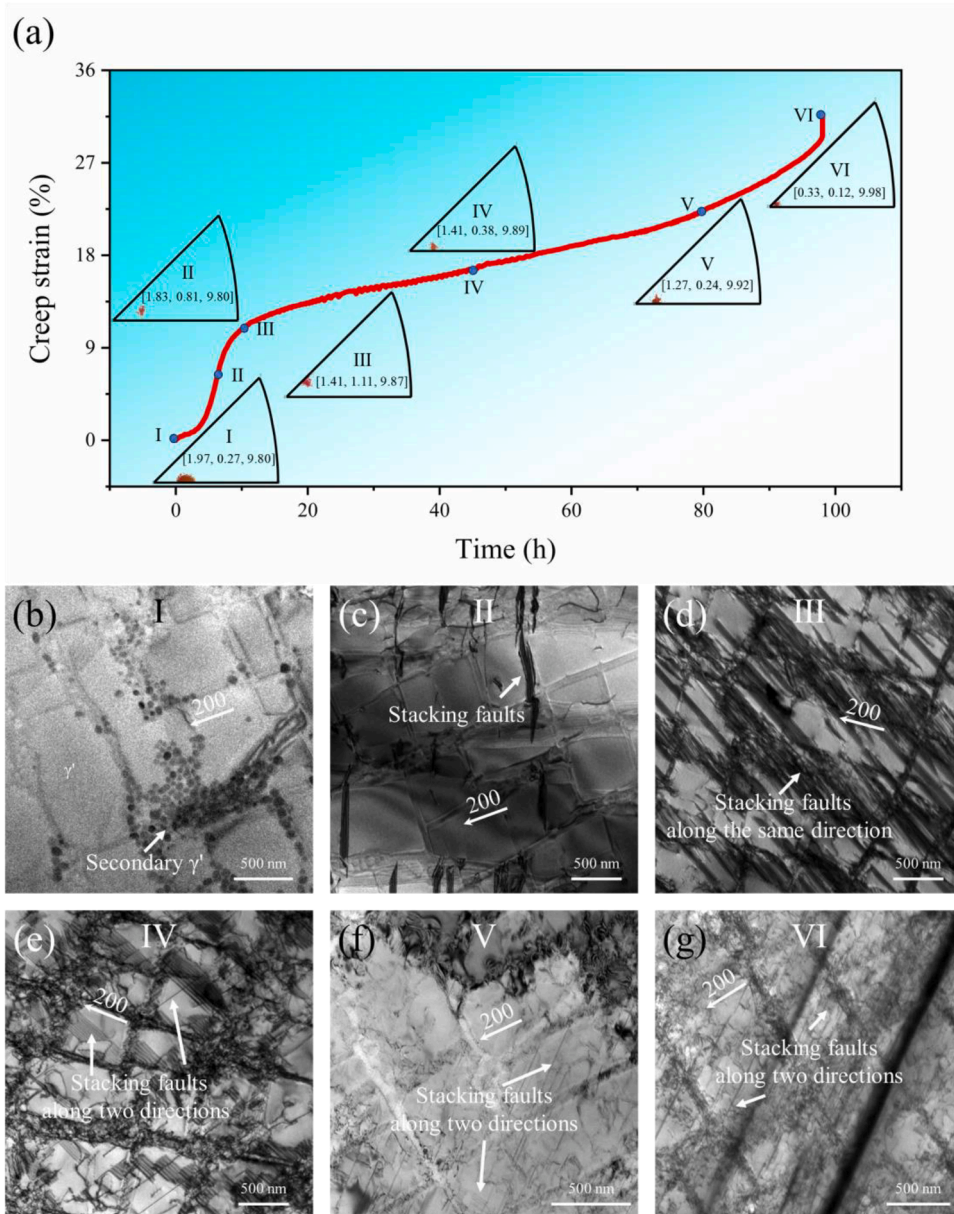


Fig. 3. (a) The creep curve of the near [001] oriented specimen under 750 $^{\circ}\text{C}$ /750 MPa, the relative position of the creep interruption experiment on the creep curve and the orientation after creep interruption experiment; (b) Microstructures of specimens at Stage I; (c) Microstructures of specimens at Stage II; (d) Microstructures of specimens at Stage III; (e) Microstructures of specimens at Stage IV; (f) Microstructures of specimens at Stage V; (g) Microstructures of specimens at Stage VI.

influence of two types of slip systems, and $\{111\}$ $\langle 110 \rangle$ slip systems should be activated before $\{111\}$ $\langle 112 \rangle$ slip systems.

Fig. 2a describes the illustration of orientation rotation process dominated by $\{111\}$ $\langle 110 \rangle$ slip systems. The $[001]$ - $[101]$ - $[111]$ standard stereographic triangle could be divided into several different zones according to the number of slip systems activated, such as single slip zone, double slip zone, and multiple slip zone. The multiple slip zone includes $[010]$, $[110]$ and $[111]$ poles, $[001]$ - $[101]$, $[101]$ - $[111]$, and $[001]$ - $[111]$ boundaries are regions of double slip, and the internal area of the triangle only has one activated slip system (Li et al., 1999a, b, c; MacKay and Maier, 1982; Sun et al., 2022). Although these boundaries all belong to the double slip zone, three sides correspond to quite different double-slip orientations with differently active slip systems and corresponding dislocation reactions during tensile deformation: i.e. the critical double slip ($[001]$ - $[101]$), the coplanar double slip ($[101]$ - $[111]$), and the conjugate double slip ($[001]$ - $[111]$) (Li et al., 1999a, b, c). For the specimen in the internal area of the triangle, as only the $(\bar{1}\bar{1}1)$ $[011]$ slip system is activated, it will first rotate toward $[011]$ direction. When rotating to the $[001]$ - $[111]$ boundary, the specimen will rotate along the $[010]$ - $[11\bar{1}]$ boundary under the action of conjugate double slip. Usually, when rotating to the position of $[112]$, the specimen no longer rotates, as the effects of the two slip systems cancel out each other (Li et al., 2022a; MacKay and Maier, 1982; Qu et al., 2024b; Ye et al., 2022). Then the illustration of orientation rotation process dominated by $\{111\}$ $\langle 112 \rangle$ slip systems is shown in Fig. 2b. The standard stereographic triangle would also be divided into three different zones. Similar to Fig. 2a, $[001]$, $[101]$ and $[111]$ poles also have multiple activated slip system, and the $[101]$ - $[111]$ boundary belongs to the coplanar double slip. Then the critical double slip includes the $[001]$ -X, X-Y and Y- $[101]$ boundaries, and the conjugate double slip includes the $[001]$ -Y and X- $[111]$ boundaries. The internal area in the standard stereographic triangle is the region of single slip, which is divided into two parts, $[001]$ -X-Y and X-Y- $[101]$ - $[111]$. In $[001]$ -X-Y region, (111) $[\bar{1}\bar{1}2]$ is the dominant active slip system, but for X-Y- $[101]$ - $[111]$ region, that is $(\bar{1}\bar{1}1)$ $[121]$ slip system (MacKay and Maier, 1982; Qu et al., 2023a, 2024a). The sample near $[001]$ orientation is located in $[010]$ -X-Y region. Hence, the specimen will first rotate to the $[001]$ -Y boundary under the action of (111) $[\bar{1}\bar{1}2]$ slip system. Then, the specimen will rotate to the $[010]$ pole along the $[010]$ -Y boundary under the action of conjugate double slip.

For samples with a certain deviation angle from the $[001]$ orientation, during the creep process at intermediate temperatures, the crystal orientation always rotates towards the $[001]$ direction, hence it cannot be caused solely by the $\{111\}$ $\langle 110 \rangle$ slip system. Therefore, three possible orientation rotation models are shown in Fig. 2c. Model A is the direct rotation model, which satisfies most of the orientation changes before and after creep rupture. Model B is the model that the rotation is dominated by $\{111\}$ $\langle 112 \rangle$ slip system. The sample will first rotate to the $[001]$ -Y ($[001]$ - $[101]$) boundary under single slip, and then rotate towards the $[001]$ direction under conjugate double slip. Model C is the model that the rotation is dominated by both $\{111\}$ $\langle 110 \rangle$ and $\{111\}$ $\langle 112 \rangle$ slip systems. In Model C, the sample first rotates towards the $[001]$ -X ($[001]$ - $[111]$) boundary under the action of $\{111\}$ $\langle 110 \rangle$ slip system. When the dislocation reactions between γ -channel dislocation families with different Burgers vectors cause the $\{111\}$ $\langle 112 \rangle$ slip system to dominate deformation, the specimen will rotate under the action of $\{111\}$ $\langle 112 \rangle$ slip system. It will also first rotate to the $[001]$ -Y boundary, and then rotate towards the $[001]$ direction. In these models, Model A cannot reflect the orientation rotation path and is also unable to explain the impact of deviation direction on alloy properties at the same deviation angle. Model B only considers the effect of $\{111\}$ $\langle 112 \rangle$ slip system. Model C comprehensively considers the influence of $\{111\}$ $\langle 110 \rangle$ and $\{111\}$ $\langle 112 \rangle$ slip systems. Therefore, Model C seems more reasonable, but further proof is needed.

4. Results

4.1. Orientation changes and microstructure evolution in different stages of creep

To reveal the orientation rotation and deformation mechanism of the near $[001]$ oriented single crystal superalloys during the creep process at intermediate temperatures and high stress, some specimens with $[1.97, 0.27, 9.80]$ orientation were adopted for creep interruption experiments at $750^\circ\text{C}/750\text{ MPa}$. These specimens were biased towards the $[001]$ - $[101]$ boundary, with a deviation angle of approximately 11.82° from the $[001]$ orientation. Fig. 3a describes the creep curve of the near $[001]$ oriented specimen at $750^\circ\text{C}/750\text{ MPa}$. The creep curve can be divided into three creep stages: primary, secondary and tertiary creep stages (Drew et al., 2004; Li et al., 2022b; Zhang et al., 2024a). According the creep curve, several specific stages were for creep interruption experiment, and the relative position were also shown in Fig. 3a. Stage I is the initial state of the creep specimen, and Stage II is the position with the highest strain rate at the primary stage. Stage III is the transition position between the primary and secondary stages. Then Stage IV represents the middle position of the secondary stage, and V is the transition position between the secondary and tertiary stages. Finally, VI is the state of the specimen after creep rupture.

Fig. 3a also shows the orientation changes of samples in different creep stages. Stage I is the initial orientation, which is $[1.97, 0.27, 9.80]$, close to the $[001]$ - $[101]$ boundary. After a short period of creep deformation, the orientation of Stage II rotates towards the $[001]$ - $[111]$ boundary, and the orientation is $[1.83, 0.81, 9.80]$. The orientation at the transition position III between the primary and secondary stages has rotated to almost the $[001]$ - $[111]$ boundary, but there is still a slight orientation deviation with the boundary. At Stage III, the crystal orientation is $[1.41, 1.11, 9.87]$. In the secondary stage, during the deformation process from III to V, the orientation would rotate from the $[001]$ - $[111]$ boundary towards the $[001]$ - $[101]$ boundary. The specimen at V is closer to the $[001]$ orientation compared to IV. Orientations of the specimen at IV and V are $[1.41, 1.11, 9.87]$ and $[1.27, 0.24, 9.92]$, and the deviations from $[001]$ orientation are 10.29° and 7.42° , respectively. After creep rupture, the orientation of VI is $[0.33, 0.12, 9.98]$, which is closer to the $[001]$ orientation, and the deviation between them is only 2.01° .

Except for orientation changes, there are also significant differences in the microstructure at different creep stages, and Figs. 3b-g shows the microstructure of the samples at different creep stages. Fig. 3b is the initial microstructure before creep, which exhibits a

dual phase γ/γ' microstructure with a uniform distribution of γ' phases in the γ matrix. In addition, there are lots of spherical secondary γ' phases, which are formed during the air cooling after primary aging and would improve mechanical properties at intermediate temperatures (Qu et al., 2020, 2023b; Xia et al., 2023). Fig. 3c is the deformation microstructure of the specimen at Stage II, in which there are a few stacking faults along the same direction. And it is also found some matrix dislocation on the surface of the specimen. Stacking faults are formed by the reaction of matrix dislocations with different Burgers vectors (Drew et al., 2004; Galindo-Nava et al., 2023; Qu et al., 2021). Hence, before the formation of stacking faults, a large number of matrix dislocations should first accumulate in the γ matrix channel. However, due to the narrow γ matrix channel and the influence of diffraction conditions, these matrix dislocations are not clearly displayed. Compared to Stage II, the microstructure at Stage III contains more stacking faults as shown in Fig. 3d. These stacking faults are parallel to each other and along the same direction, indicating that only a $\{111\} \langle 112 \rangle$ slip system has been

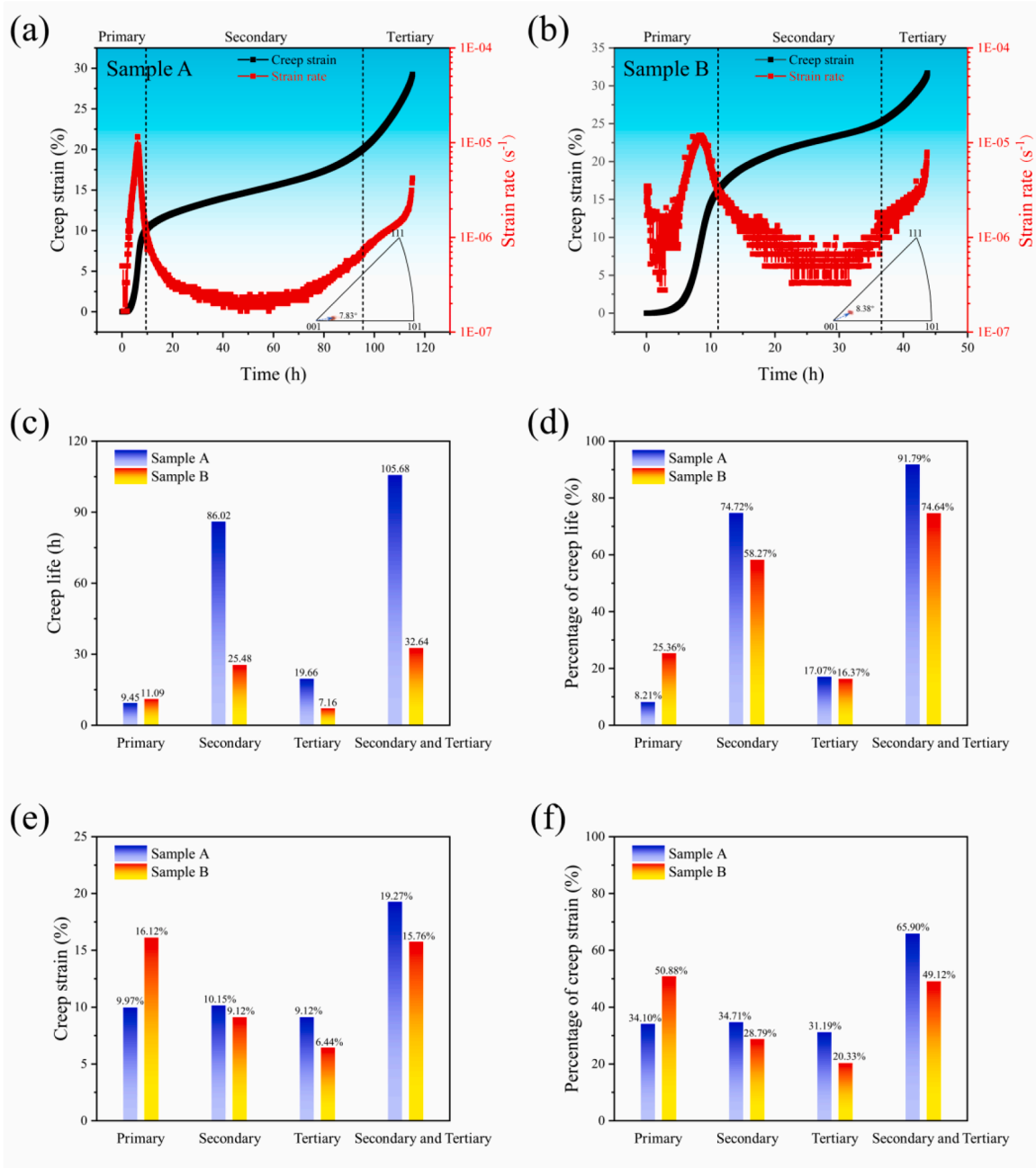


Fig. 4. (a) The creep curve and the initial orientation of sample A; (b) The creep curve and the initial orientation of sample B; (c) Creep life of different samples at different stages; (d) Percentage of creep life of different samples at different stages; (e) Creep strain of different samples at different stages; (f) Percentage of creep strain of different samples at different stages.

activated (Qu et al., 2021). At stage IV shown in Fig. 3e, the deformation microstructure contains stacking faults along two orientations, and there are more stacking faults in one direction, while there are relatively fewer stacking faults in the other direction. Therefore, the second $\{111\} \langle 112 \rangle$ slip system has been activated, but the number is relatively less. For stages V and VI, the density of stacking faults sharply increases with the increase of creep strain. As shown in Figs. 3f and 3g, there are multiple stacking faults in a γ' phase, which is completely different from the early creep stage. The results in **Supplementary Materials** further demonstrate that as the strain increases, the number of activated $\{111\} \langle 112 \rangle$ slip systems also increases, which is consistent with the increase in stacking fault density.

4.2. Effect of orientation deviation from precise [001] on creep properties

In order to further elucidate the effect of orientation deviation from precise [001] on creep properties at intermediate temperatures, two specimens with similar orientation deviation angles but different orientation deviation directions were used for creep testing to at 750 °C/750 MPa. As shown in Fig. 4a, the orientation of sample A is close to the [001]-[101] boundary and has a deviation angle of 7.83° from precise [001] orientation. Its creep life is approximately 115.13 h, and the creep curve shows a typical three-stage creep. For sample B, the deviation angle from precise [001] is similar to sample A, approximately 8.38°, as shown in Fig. 4b. Different from sample A, the orientation of sample B is close to the [001]-[111] boundary, and its creep life is only 43.73 h. The minimum strain rate of sample A is about $1.66 \times 10^{-7} \text{ s}^{-1}$, while that of sample B is approximately twice that of sample A, which is $3.32 \times 10^{-7} \text{ s}^{-1}$.

As shown in Fig. 4, there are significant differences in the creep life and creep behavior between two specimens with similar orientation deviation angles but different deviation directions. Although both specimens exhibit typical three-stage creep behavior, there are significant differences in the creep life and creep strain at different stages. Similar to previous research, the sample near [001]-[111] boundary exhibits a large primary creep strain (MacKay and Maier, 1982; Rae and Reed, 2007; Rae et al., 2000). Fig. 4c-f compares some creep parameters of different stages, which can better reveal the effect of orientation deviation on different creep stages. Fig. 4c shows the creep life of two samples at different creep stages. At the primary stage, the creep life of sample A is 9.45 h, while sample B has a longer creep life of 11.09 h. The creep life of sample A both at the secondary and the tertiary stages is much longer than that of sample B. At the secondary stage, the creep life of sample A is as high as 86.02 h, while sample B is only 25.48 h. At the tertiary stage, the creep lives of samples A and B are 19.66 h and 7.16 h, respectively. Fig. 4d further compares the percentage of creep life at different stages. The results show that the percentage of creep life at the primary stage of sample B is as high as 25.36 %, while sample A only has 8.21 %. At the secondary and the tertiary stages, the percentage of creep life of sample B is relatively low, at 58.27 % and 16.37 %, respectively, while sample A is 74.72 % and 16.37 %, respectively. For sample A, the sum of creep life at the secondary and the tertiary stages is 105.68 h, accounting for 91.79 % of the total creep life. For sample B, the sum of creep life at the secondary and the tertiary stages is 32.64 h, accounting for 74.64 % of the total creep life.

In addition, the two samples exhibit significant differences in creep strain at different creep stages. As shown in Figs. 4e and 4f, the strain of sample B at the primary stage is 16.12 %, accounting for 50.88 % of the total strain, while the strain of sample A is relatively small at 9.97 %, accounting for approximately 34.10 %. At the secondary stage, the strain and its percentage to the total strain of specimen B are lower than those of specimen A. The strain and its percentage of specimen B are 9.12 % and 28.79 %, respectively, while the strain and its percentage of specimen A are 10.15 % and 34.71 %, respectively. Same as the secondary stage, the strain and its percentage of sample A at the tertiary stage are higher than those of sample B. The strain of samples A and B are 9.12 % and 6.44 %, respectively, and the percentage of creep strain is 31.19 % and 20.33 %. For sample A, the sum of creep strain at the secondary and the tertiary stages is 19.27 %, accounting for 65.90 %. For sample B, the sum of creep strain at the secondary and the tertiary stages is 15.76 %, accounting for 49.12 %.

5. Discussions

5.1. Orientation rotation path during creep at intermediate temperatures

The orientation changes in Fig. 3a indicates that the crystal orientation rotates from the [001]-[101] boundary towards the [001]-[111] boundary at the primary stage (from I to III), which is consistent with the orientation rotation law dominated by the $\{111\} \langle 110 \rangle$ slip system shown in Fig. 2a. Although, some stacking faults have already appeared in the microstructure of the specimens at the primary stage, as shown in Fig. 3c. However, the number of stacking faults is relatively less and has not reached the threshold that can cause orientation rotation. Recently, Qu et al. (2024b) also found that although there were some stacking faults, the dominant mechanism still was the movement of matrix dislocations during tensile deformation at intermediate temperatures. The result of Heep et al. (2022) also indicated that there were numerous matrix dislocations the early stage of creep. Therefore, the orientation rotation at the primary stage is dominated by $\{111\} \langle 110 \rangle$ slip system.

At the secondary stage (from III to V) shown in Fig. 3a, the crystal orientation rotates from the [001]-[111] boundary towards the [001]-[101] boundary. At the tertiary stage, the crystal orientation rotates towards the [001] pole along the [001]-[101] boundary until the specimen fractures. The rotation path of the specimen at the secondary and tertiary stages is consistent with the orientation rotation law dominated by the $\{111\} \langle 112 \rangle$ slip system shown in Fig. 2b, and the orientation rotation at the secondary stage is dominated by $\{111\} \langle 112 \rangle$ single slip, while it is dominated by $\{111\} \langle 112 \rangle$ double slip at the tertiary stage. Although there are stacking faults along two orientations at the secondary stage, the density of stacking faults in the two directions is different. There are more stacking faults formed by the movement of the first activated $\{111\} \langle 112 \rangle$ slip system, which is also the dominant slip system during the secondary stage. Then, there are fewer stacking faults formed by the movement of the second activated $\{111\} \langle 112 \rangle$ slip

system, which has not reached the threshold that can cause orientation rotation during the secondary stage and will result in orientation rotation during the tertiary creep stage under the joint action with the first activated slip system. To sum up, the orientation rotation is caused by the combined action of the $\{111\} \langle 110 \rangle$ slip system and the $\{111\} \langle 112 \rangle$ slip system during the creep process at intermediate temperatures. The $\{111\} \langle 110 \rangle$ slip system dominates the rotation process at the primary stage, the $\{111\} \langle 112 \rangle$ single slip results in the orientation rotation at the secondary stage, and the $\{111\} \langle 112 \rangle$ conjugate double slip causes the rotation process at the tertiary stage.

Fig. 5 describes the orientation rotation paths at different positions under the same orientation deviation degree with [001] pole. In Fig. 5a, the initial orientation of the sample is on the [001]-Y ([001]-[101]) boundary. At the primary stage, the orientation will first rotate towards the boundary of [001]-X ([001]-[111]) under the action of $\{111\} \langle 110 \rangle$ single slip, as shown by the blue solid line arrow in Fig. 5a. When the orientation rotates to the middle region between the boundaries of [001]-Y and [001]-X, a large number of matrix dislocations react with each other, causing the $\{111\} \langle 112 \rangle$ slip system to dominate. At this time, the secondary stage does begin and the orientation will rotate towards the [001]-Y boundary under the action of $\{111\} \langle 112 \rangle$ single slip, as shown by the red solid line arrow in Fig. 5a. Subsequently, after the orientation rotates to the [001]-Y boundary, the $\{111\} \langle 112 \rangle$ double slip will cause the specimen orientation to rotate towards the [001] pole until the specimen fractures, as shown by the yellow solid line arrow in Fig. 5a. When the initial orientation of the sample is on the middle region between the boundaries of [001]-Y and [001]-X, the orientation rotation path slightly different. As shown in Fig. 5b, the rotation process at the primary stage is divided into two steps. First, the orientation will rotate towards the [001]-X boundary under the action of $\{111\} \langle 110 \rangle$ single slip. Then, the orientation will rotate towards [111] direction along the [001]-X boundary, as shown by the blue dashed line arrow in Fig. 5b. When the $\{111\} \langle 112 \rangle$ slip system dominates, the specimen will rotate successively under the action of $\{111\} \langle 112 \rangle$ single slip and $\{111\} \langle 112 \rangle$ conjugate double slip, until the specimen fractures. Fig. 5c describes the orientation rotation path of the sample located in the [001]-X boundary. Because the [001]-X boundary belongs to the conjugate double slip zone of the $\{111\} \langle 110 \rangle$ slip system, the sample is under the conjugate double slip action at the beginning of creep. Therefore, at the primary stage, the specimen rotates along the direction of the blue dashed line under the conjugate double slip action of $\{111\} \langle 110 \rangle$, and then undergoes rotation under the action of the $\{111\} \langle 112 \rangle$ slip system.

At the primary stage, the sample near the [001]-X boundary will rotate further away from the [001] pole and [001]-Y boundary under the action of $\{111\} \langle 110 \rangle$ slip system. Then, the sample need large rotation to reach the [001]-Y boundary at the secondary stage to produce $\{111\} \langle 112 \rangle$ double slip. The increased rotation degree at the secondary stage can exacerbate nonuniform deformation and lead to premature fracture (Gupta et al., 2023; Qu et al., 2023a). Therefore, the crystal orientation during the tertiary stage cannot rotate to the [001] pole. As shown in Fig. 5, the closer the initial orientation is to the [001]-X boundary, the further away the orientation after the creep rupture is from the [001] pole.

5.2. Orientation sensitive mechanism of creep life at intermediate temperatures

The results in Fig. 4 further indicates the percentage of creep life at the primary stage of the samples near the [001]-[111] boundary is much larger than those close to the [001]-[101] boundary. Furthermore, the samples near the [001]-[111] boundary also exhibit a large primary creep strain and strain percentage. Therefore, $\{111\} \langle 110 \rangle$ slip system has a greater effect on the primary stage of samples near the [001]-[111] boundary. The activity of $\{111\} \langle 110 \rangle$ slip system is influenced by Schmid factors. And the standard stereographic projection is beneficial for revealing the relationship between crystal orientation and the activity of slip systems. Fig. 6a is the standard [001] stereographic projection of the cubic crystal system, and the specimen is located at the [001]-[101]-[111]

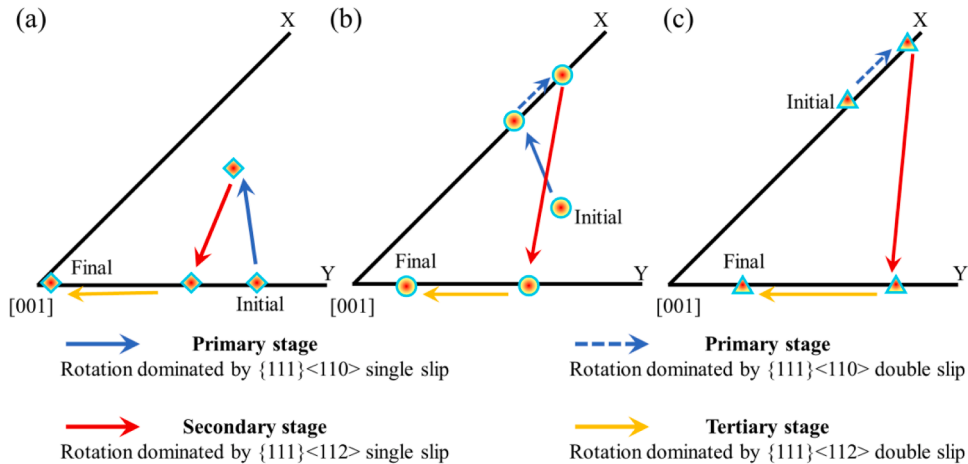


Fig. 5. Orientation rotation paths at different positions under the same orientation deviation degree. (a) The orientation path of the sample on the [001]-Y ([001]-[101]) boundary; (b) The orientation path of the sample on the middle region between the boundaries of [001]-Y and [001]-X; (c) The orientation path of the sample on the [001]-X ([001]-[111]) boundary.

standard stereographic triangle (Label α). Therefore, only the $(\bar{1}\bar{1}1)[011]$ slip system is activated in the internal area of the α triangle, which causes the orientation to rotate towards the $[001]$ - $[111]$ boundary. Samples located on the $[001]$ - $[111]$ boundary, is located at the junction of the α triangle and the β triangle. Therefore, the $\{111\}\langle 110\rangle$ conjugate double slip systems that can be activated are $(\bar{1}\bar{1}1)[011]$ and $(\bar{1}\bar{1}1)[101]$. Figs. 6b and 6c are the Schmid factors for $(\bar{1}\bar{1}1)[011]$ and $(\bar{1}\bar{1}1)[101]$ slip systems of samples with different orientations, respectively. The black dashed line indicates a deviation of 15° from precise $[001]$ orientation. In the entire region within 15° of deviation from $[001]$ orientation, the Schmid factors for $(\bar{1}\bar{1}1)[011]$ slip system have large values, which are easily activated. For $(\bar{1}\bar{1}1)[101]$ slip system, the samples near the $[001]$ - $[111]$ boundary have larger Schmid factors, while Schmid factors of the samples close to the $[001]$ - $[111]$ boundary are relatively small. Therefore, for samples near the $[001]$ - $[111]$ boundary, they will be simultaneously affected by two $\{111\}\langle 110\rangle$ slip systems with larger Schmid factors at the primary stage, resulting in a larger primary strain.

Although the primary creep is dominated by $\{111\}\langle 110\rangle$ slip system, it is also affected by $\{111\}\langle 112\rangle$ slip system. Because

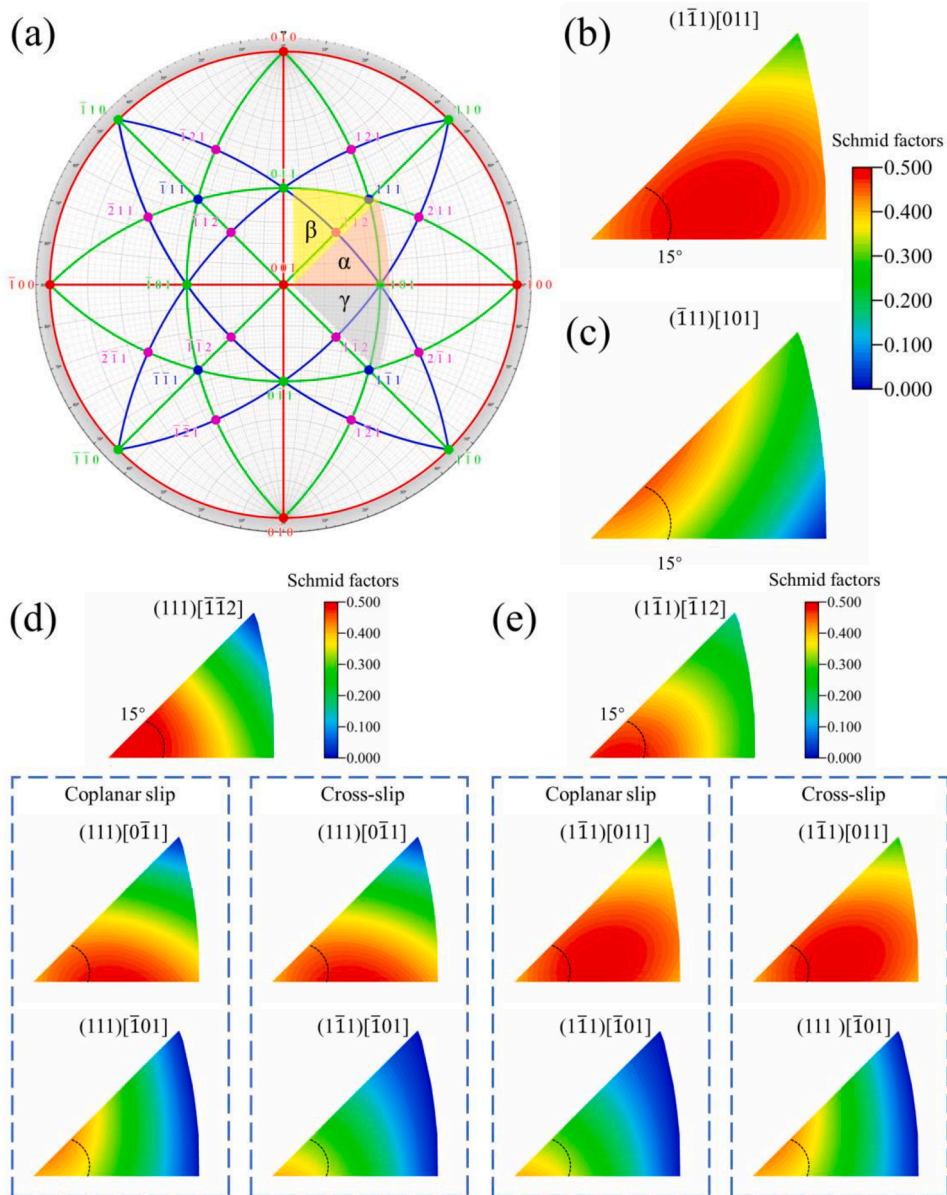


Fig. 6. (a) The standard $[001]$ stereographic projection; (b) Schmid factors for $(\bar{1}\bar{1}1)[011]$ slip system of samples with different orientation; (c) Schmid factors for $(\bar{1}\bar{1}1)[101]$ slip system of samples with different orientation. (d) Schmid factors for $(111)[\bar{1}\bar{1}2]$ slip system and $\{111\}\langle 110\rangle$ slip systems that caused the activity of $(111)[\bar{1}\bar{1}2]$ slip system of samples with different orientation; (e) Schmid factors for $(\bar{1}\bar{1}1)[\bar{1}\bar{1}2]$ slip system and $\{111\}\langle 110\rangle$ slip systems that caused the activity of $(\bar{1}\bar{1}1)[\bar{1}\bar{1}2]$ slip system of samples with different orientation.

secondary creep does not begin until $\{111\} \langle 112 \rangle$ slip system can dominate orientation rotation. The activity of the $\{111\} \langle 112 \rangle$ slip system requires different $\{111\} \langle 110 \rangle$ slip systems to react with each other (Qu et al., 2021; Rae and Reed, 2007). As shown in Fig. 6a, the sample is located in the α triangle and near the $[001]$ orientation, hence the $\{111\} \langle 112 \rangle$ slip system that causes the orientation rotation is $(111) [\bar{1}\bar{1}2]$. The $[001]$ - $[101]$ boundary is the junction the α triangle and the γ triangle, the sample near the $[001]$ orientation can rotate towards $[001]$ pole the action of conjugate double slip of $(111) [\bar{1}\bar{1}2]$ and $(\bar{1}\bar{1}1) [\bar{1}12]$. Fig. 6d shows the Schmid factors for $(111) [\bar{1}\bar{1}2]$ slip system of samples with different orientations and the Schmid factors for $\{111\} \langle 110 \rangle$ slip systems that cause the activity of $(111) [\bar{1}\bar{1}2]$ slip system. There are two models of the activity of $\{111\} \langle 112 \rangle$ slip system: the coplanar slip model and the cross-slip model (León-Cázares et al., 2022; Long et al., 2017; Qu et al., 2024a; Sass and Feller-Kniepmeier, 1998; Wu et al., 2018). Both models are also depicted in Fig. 6d. In the whole region within 15° of the deviation from $[001]$ orientation, the Schmid factors for $(111) [\bar{1}\bar{1}2]$ are similar, however, there are great differences in the Schmid factors for $\{111\} \langle 110 \rangle$ slip systems that cause the activity of $(111) [\bar{1}\bar{1}2]$ slip system. Whether in the coplanar slip or the cross-slip models, $(111) [0\bar{1}1]$ slip system is crucial. It is obvious that the Schmid factor for $(111) [0\bar{1}1]$ slip system of the sample near the $[001]$ - $[101]$ boundary is higher than that of the sample near $[001]$ - $[111]$ boundary. Therefore, the sample near the $[001]$ - $[111]$ boundary needs a larger amount of deformation to form enough $(111) [0\bar{1}1]$ slip system caused the activity of $(111) [\bar{1}\bar{1}2]$ slip system. For $(111) [\bar{1}\bar{1}2]$ slip system in the coplanar slip model, the Schmid factor in the entire region within 15° deviation from the $[001]$ orientation is not significantly different. For $(\bar{1}\bar{1}1) [\bar{1}12]$ slip system in the cross-slip model, the Schmid factor of the sample near the $[001]$ - $[101]$ boundary is also higher than that of the sample near $[001]$ - $[111]$ boundary. Therefore, the sample near the $[001]$ - $[111]$ boundary also needs a larger amount of deformation to form enough $(\bar{1}\bar{1}1) [\bar{1}12]$ slip system caused the activity of $(111) [\bar{1}\bar{1}2]$ slip system.

To sum up, there are three factors that result in the sample near the $[001]$ - $[111]$ boundary exhibiting a large primary strain and a large orientation rotation. Firstly, the $\{111\} \langle 110 \rangle$ slip system that causes its rotation has a large Schmid factor. Secondly, the activity of $\{111\} \langle 110 \rangle$ conjugate double slip can intensify the deformation and orientation rotation of the specimen. Finally, the Schmid factors for $\{111\} \langle 110 \rangle$ slip systems that cause the activity of $(111) [\bar{1}\bar{1}2]$ slip system is small. Then the sample requires greater strain to form sufficient matrix dislocations, which is also needed by the activity of $(111) [\bar{1}\bar{1}2]$ slip system. Only the activity of $(111) [\bar{1}\bar{1}2]$ slip system meets the threshold that can cause orientation rotation, secondary creep can begin. Otherwise, the primary strain and orientation rotation will still increase.

At the primary stage, the sample near $[001]$ - $[111]$ boundary exhibits a large primary strain, which causes the orientation to rotate further away from the $[001]$ pole and the $[001]$ - $[101]$ boundary. Therefore, at the secondary stage, the sample requires a large rotation to reach the $[001]$ - $[101]$ boundary. The large rotation will exacerbate the asynchrony of rotation, which would result in nonuniform deformation and local stress concentration, and exacerbate the failure of the specimen (Gupta et al., 2023; Qu et al., 2023a). At the tertiary stage, the sample rotates towards to $[001]$ pole under the action of conjugate double slip of $(111) [\bar{1}\bar{1}2]$ and $(\bar{1}\bar{1}1) [\bar{1}12]$ slip systems. For $(111) [\bar{1}\bar{1}2]$ slip system, it has been extensively activated at the secondary stage. For $(\bar{1}\bar{1}1) [\bar{1}12]$ slip system, it is starting to be activated at the secondary stage and reaches its threshold that can cause orientation rotation at the tertiary stage. Fig. 6e shows the Schmid factors for $(\bar{1}\bar{1}1) [\bar{1}12]$ slip system of samples with different orientations and the Schmid factors for $\{111\} \langle 110 \rangle$ slip systems that cause the activity of $(\bar{1}\bar{1}1) [\bar{1}12]$ slip system. In the entire region within 15° of deviation from $[001]$ orientation, the

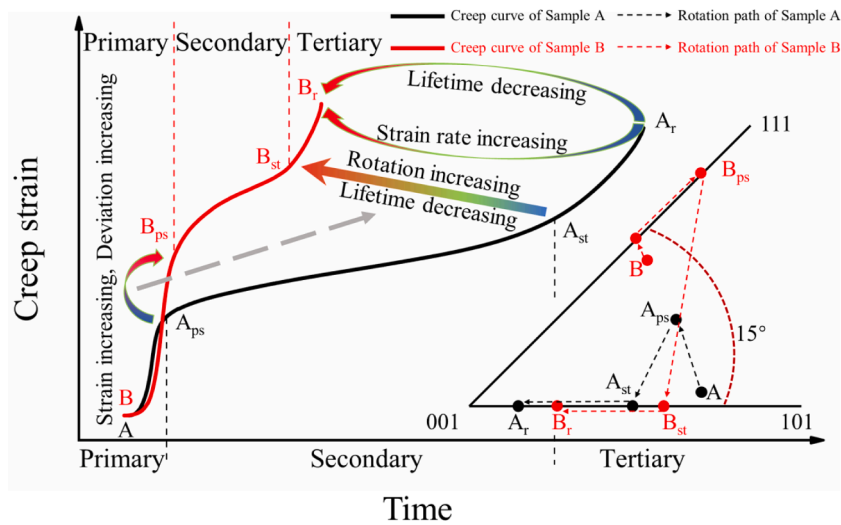


Fig. 7. The mechanism of the effect of orientation deviation on alloy properties. Schematic illustration showing three creep stages and orientation rotation paths for alloys with different orientations. Samples A and B have similar deviation angles from precise $[001]$, with sample A approaching the $[001]$ - $[101]$ boundary and sample B near the $[001]$ - $[111]$ boundary. The junctions between primary/secondary stages and secondary/tertiary stages for sample A and sample B are denoted as A_{ps} , A_{st} , B_{ps} and B_{st} , respectively. A_r and B_r represent the state of samples A and B after creep rupture, respectively.

Schmid factors for $(\bar{1}\bar{1}1)$ $[\bar{1}12]$ slip system have large values, especially for the sample near $[001]$ - $[101]$ boundary. For $\{111\}$ $\langle 110 \rangle$ slip systems that cause the activity of $(\bar{1}\bar{1}1)$ $[\bar{1}12]$ slip system, the $(\bar{1}\bar{1}1)$ $[011]$ slip system has large Schmid factors and the matrix dislocations formed by the activity of $(\bar{1}\bar{1}1)$ $[011]$ slip system accumulate continuously at the primary stage. The $(\bar{1}\bar{1}1)$ $[\bar{1}01]$ slip system in the coplanar slip model and the (111) $[\bar{1}01]$ slip system in the cross-slip model are activated and the corresponding matrix dislocation also accumulate continuously during the continuous deformation process. Therefore, the sample shows a large creep strain rate at the tertiary stage.

Fig. 7 illustrates the mechanism of the effect of orientation deviation from precise $[001]$ on creep at intermediate temperature. In the Figure, samples A and B have similar deviation angles from precise $[001]$, with sample A approaching the $[001]$ - $[101]$ boundary and sample B near the $[001]$ - $[101]$ boundary. The junctions between primary/secondary stages and secondary/tertiary stages for sample A and sample B are denoted as A_{ps} , A_{st} , B_{ps} and B_{st} , respectively. A_r and B_r represent the state of samples A and B after creep rupture, respectively. At the primary stage, the deformation is mainly dominated by the $\{111\}$ $\langle 110 \rangle$ slip system. The orientation of sample A will rotate to the middle location (A_{ps}) of the two boundaries under the action of $\{111\}$ $\langle 110 \rangle$ single slip action. For sample B, it will first rotate to the $[001]$ - $[111]$ boundary under the action of single slip and then rotate to the B_{ps} under the action of conjugate double slip. Due to the relatively small Schmid factor of the $\{111\}$ $\langle 110 \rangle$ slip systems that can form the $\{111\}$ $\langle 112 \rangle$ slip system at sample B, it needs to accumulate significant deformation to generate a sufficient number of matrix dislocations, therefore the degree of orientation rotation is relatively large. Meanwhile, sample B is near the $[001]$ - $[111]$ boundary, and the $\{111\}$ $\langle 110 \rangle$ conjugate double slip will also increase the deformation of sample B, as well as the degree of orientation rotation. Therefore, as shown in Fig. 7, sample B exhibits a large primary creep strain and a large orientation rotation at the end of the primary stage.

At the secondary stage, the main deformation mechanism is dominated by the single slip of $\{111\}$ $\langle 112 \rangle$ slip system, and the orientation will rotate towards the $[001]$ - $[101]$ boundary. Due to the large creep strain and rotation of sample B at the primary stage, the position of B_{ps} is further away from the $[001]$ - $[101]$ boundary compared to the position of A_{ps} . Therefore, sample B requires large rotations to rotate to the $[001]$ - $[101]$ boundary, to produce $\{111\}$ $\langle 112 \rangle$ conjugate double slip and the onset of the tertiary stage. The large rotation can accelerate the fracture of the sample and reducing the creep life. At the tertiary stage, the $\{111\}$ $\langle 112 \rangle$ conjugate double slip will cause the orientation to rotate along the $[001]$ - $[101]$ boundary towards the $[001]$ pole. However, due to the large rotation of sample B in the first two stages, the local deformation is more severe, causing premature fracture before it rotates to the $[001]$ pole. As shown in Fig. 7, the orientation after creep rupture of sample A (A_r) is closer to the $[001]$ orientation than that of sample B (B_r).

6. Conclusion

In summary, the orientation sensitivity of creep life in near $[001]$ oriented Ni-based single crystal superalloys at intermediate temperatures has been unveiled in this work. The key findings can be summarized as follows:

- (1) During the creep process at intermediate temperature, the orientation rotation path indicates the primary stage is dominated by $\{111\}$ $\langle 110 \rangle$ slip system, while both the secondary and the tertiary stages are dominated by $\{111\}$ $\langle 112 \rangle$ slip system. In addition, the secondary stage is dominated by $\{111\}$ $\langle 112 \rangle$ single slip, while the tertiary stage is dominated by $\{111\}$ $\langle 112 \rangle$ conjugate double slip.
- (2) The creep life exhibits obvious orientation sensitivity in the vicinity of $[001]$ orientation, and the creep life of samples near the $[001]$ - $[101]$ boundary was much longer than those close to the $[001]$ - $[111]$ boundary. For samples near the $[001]$ - $[101]$ boundary, the proportion of primary creep life to the total creep life is relatively small and it also has a low primary creep strain and strain proportion. In contrast, the sample near the $[001]$ - $[111]$ boundary exhibits a large primary creep strain and strain proportion. The percentage of creep life at the primary stage is also much larger than that of samples near the $[001]$ - $[101]$ boundary.
- (3) The reason for the orientation sensitivity of creep life near the $[001]$ orientation is as follows: firstly, at the primary stage, the sample near the $[001]$ - $[111]$ boundary require a large deformation to generate sufficient matrix dislocations. Only a large number of matrix dislocations can react with each other to reach the threshold that can cause orientation rotation in the $\{111\}$ $\langle 112 \rangle$ single slip system, and then the secondary stage can begin. At the same time, the samples near the $[001]$ - $[111]$ boundary are subjected to $\{111\}$ $\langle 110 \rangle$ double slip resulting in significant deformation and orientation rotation further away from the $[001]$ pole and $[001]$ - $[101]$ boundary. Secondly, a large creep strain and rotation at the primary stage will require a larger rotation angle for the sample to reach the $[001]$ - $[101]$ boundary at the secondary stage, thereby initiating the double slip of the $\{111\}$ $\langle 112 \rangle$ slip system. Finally, excessive rotation can lead to greater uneven deformation and exacerbate local stress concentration in the specimen, ultimately resulting in a poorer creep life of the specimen.
- (4) The deviation angle of 15° from precise $[001]$ is not an appropriate indicator to determine whether the blade is qualified. The deviation angle between the blade stacking axis (Z) and the $[001]$ orientation not only needs to be controlled, but also the deviation direction needs to be controlled. A deviation angle of 15° from $[001]$ in $[101]$ directions can be tolerated, while a deviation towards the $[111]$ direction cannot be accepted.

CRediT authorship contribution statement

Pengfei Qu: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal

analysis, Data curation, Conceptualization. **Wenchao Yang:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Qiang Wang:** Validation, Investigation, Formal analysis, Data curation. **Chen Liu:** Validation, Software, Data curation. **Jiarun Qin:** Software, Data curation. **Jun Zhang:** Supervision, Resources, Funding acquisition. **Lin Liu:** Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Acknowledgment

This work was funded by the National Natural Science Foundation of China (52322410, 52071263, 52031012, 51771148); Self-innovation Special Fund Project of Aero Engine Corporation of China (ZZCX-2022-040); Science Center for Gas Turbine Project (P2021-A-IV-001-001); The Key Research and Development program of Shaanxi Province (2023-YBGY-432); Innovation Foundation for Doctor Dissertation of Northwestern Polytechnical University, China (CX2021057).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ijplas.2024.104035](https://doi.org/10.1016/j.ijplas.2024.104035).

References

- Darolia, R., 2019. Development of strong, oxidation and corrosion resistant nickel-based superalloys: critical review of challenges, progress and prospects. *Int. Mater. Rev.* 64, 355–380.
- Drew, G.L., Reed, R.C., Kakehi, K., Rae, C.M.F., 2004. Single crystal superalloys: the transition from primary to secondary creep. *Superalloys 2004*, 127–136.
- Dye, D., Ma, A., Reed, R., 2008. Numerical modelling of creep deformation in a CMSX-4 single crystal superalloy turbine blade. *Superalloy 2008*, 911–919.
- Fuchs, G.E., 2001. Solution heat treatment response of a third generation single crystal Ni-base superalloy. *Mater. Sci. Eng. A* 300, 52–60.
- Galindo-Nava, E.I., Schlütter, R., Messé, O.M.D.M., Argyrakis, C., Rae, C.M.F., 2023. A model for dislocation creep in polycrystalline Ni-base superalloys at intermediate temperatures. *Int. J. Plast.* 169, 103729.
- Ghosh, R.N., Curtis, R.V., McLean, M., 1990. Creep deformation of single crystal superalloys—Modelling the crystallographic anisotropy. *Acta Metallurgica et Materialia* 38, 1977–1992.
- Gupta, R., Sharma, V., Prasad, M.J.N.V., Pant, P., 2023. Orientation anisotropy and strain localization during elevated temperature tensile deformation of single crystal and bi-crystal Ni-based GTD444 superalloy. *Mater. Sci. Eng. A* 880, 145310.
- Heep, L., Bürger, D., Bonnekoh, C., Wollgramm, P., Dlouhy, A., Eggeler, G., 2022. The effect of deviations from precise [001] tensile direction on creep of Ni-base single crystal superalloys. *Scripta Mater.* 207, 114274.
- Kakehi, K., Sakaki, T., Gui, J.M., Misaki, Y., 1993. The influence of orientation and heat treatment on creep strength of single crystals of a Ni-base superalloy. In: *Proceedings of the of CFEMS*, 5. Institute of Materials, London, pp. 221–230.
- Kim, Y.-K., Kim, D., Kim, H.-K., Oh, C.-S., Lee, B.-J., 2016. An intermediate temperature creep model for Ni-based superalloys. *Int. J. Plast.* 79, 153–175.
- León-Cázares, F.D., Schlütter, R., Monni, F., Hardy, M.C., Rae, C.M.F., 2022. Nucleation of superlattice intrinsic stacking faults via cross-slip in nickel-based superalloys. *Acta Mater.* 241, 118372.
- Leverant, G.R., Kear, B.H., 1970. The mechanism of creep in gamma prime precipitation-hardened nickel-base alloys at intermediate temperatures. *Metallurg. Trans.* 1, 491–498.
- Li, L., Chen, Z., Kuroiwa, S., Ito, M., Kishida, K., Inui, H., George, E.P., 2022a. Tensile and compressive plastic deformation behavior of medium-entropy Cr-Co-Ni single crystals from cryogenic to elevated temperatures. *Int. J. Plast.* 148, 103144.
- Li, X.W., Wang, Z.G., Li, S.X., 1999a. Cyclic deformation behavior of double-slip-oriented copper single crystals I: coplanar double slip orientation on 011-111 side of the stereographic triangle. *Mater. Sci. Eng. A* 260, 132–138.
- Li, X.W., Wang, Z.G., Li, S.X., 1999b. Cyclic deformation behavior of double-slip-oriented copper single crystals III: conjugate double slip orientation on 001-111 side of the stereographic triangle. *Mater. Sci. Eng. A* 269, 166–174.
- Li, X.W., Wang, Z.G., Li, S.X., 1999c. Cyclic deformation behavior of double-slip-oriented copper single crystals: II. Critical double slip orientation on 001/011 side of the stereographic triangle. *Mater. Sci. Eng. A* 265, 18–24.
- Li, Y., Wang, L., He, Y., Zheng, W., Lou, L., Zhang, J., 2022b. Role of interfacial dislocation networks during secondary creep at elevated temperatures in a single crystal Ni-based superalloy. *Scripta Mater.* 217, 114769.
- Li, Y., Wang, L., Zhao, S., Zhang, G., Lou, L., 2022c. Creep anisotropy of a 3rd generation nickel-base single crystal superalloy in the vicinity of [001] orientation. *Mater. Sci. Eng. A* 848, 143479.
- Long, H., Liu, Y., Kong, D., Wei, H., Chen, Y., Mao, S., 2017. Shearing mechanisms of stacking fault and anti-phase-boundary forming dislocation pairs in the γ' phase in Ni-based single crystal superalloy. *J. Alloys Compd.* 724, 287–295.
- MacKay, R.A., Maier, R.D., 1982. The influence of orientation on the stress rupture properties of nickel-base superalloy single crystals. *Metall. Trans. A* 13, 1747–1754.
- Mackay, R.A., Maier, R.D., Dreshfield, R.L., 1980. Anisotropy of nickel-base superalloy single crystals. *Superalloys 1980*, 385–394.
- MacLachlan, D.W., Wright, L.W., Gunturi, S., Knowles, D.M., 2001. Constitutive modelling of anisotropic creep deformation in single crystal blade alloys SRR99 and CMSX-4. *Int. J. Plast.* 17, 441–467.

- Mukherjee, S., Kumar Kar, S., Sivaprasad, S., Tarafder, S., Viswanathan, G.B., Fraser, H.L., 2022. Creep-Fatigue Response, failure mode and deformation mechanism of HAYNES 282 Ni based superalloy: effect of dwell position and time. *Int. J. Fatigue* 159, 106820.
- Qu, P., Yang, W., Liu, C., Qin, J., Su, H., Zhang, J., Liu, L., 2023a. The asynchrony of orientation rotation at 750°C/750 MPa creep in a [011]-oriented Ni-based single crystal superalloy. *Mater. Sci. Eng. A* 869, 144823.
- Qu, P., Yang, W., Liu, C., Qin, J., Wang, Q., Zhang, J., Liu, L., 2024a. Creep anisotropy dominated by orientation rotation in Ni-based single crystal superalloys at 750°C/750 MPa. *J. Mater. Sci. Technol.* 186, 91–103.
- Qu, P., Yang, W., Liu, C., Qin, J., Wang, Q., Zhang, J., Liu, L., 2024b. Tensile deformation dominated by matrix dislocations at intermediate temperatures revealed using in-situ EBSD in superalloys. *Mater. Res. Lett.* 12, 116–124.
- Qu, P., Yang, W., Qin, J., Cao, K., Liu, C., Zhang, J., Liu, L., 2021. Reveal the stacking fault shearing mechanism by its annihilation process in Ni-based single crystal superalloys. *Mater. Charact.* 180, 111419.
- Qu, P., Yang, W., Qin, J., Liu, C., Cao, K., Zhang, J., Liu, L., 2020. Precipitation behavior and chemical composition of secondary γ' precipitates in a Re-containing Ni-based single crystal superalloy. *Intermetallics* 119, 106725.
- Qu, P., Yang, W., Qin, J., Liu, C., Duan, X., Zhang, J., Liu, L., 2023b. Determining the precipitation temperature of secondary γ' phases by remelting method in Ni-based single crystal superalloys. *Intermetallics* 162, 108017.
- Rae, C.M.F., Matan, N., Reed, R.C., 2001. The role of stacking fault shear in the primary creep of [001]-oriented single crystal superalloys at 750°C and 750 MPa. *Mater. Sci. Eng. A* 300, 125–134.
- Rae, C.M.F., Reed, R.C., 2007. Primary creep in single crystal superalloys: origins, mechanisms and effects. *Acta Mater.* 55, 1067–1081.
- Rae, C.M.F., Rist, M.A., Cox, D.C., Reed, R.C., Matan, N., 2000. On the primary creep of CMSX-4 superalloy single crystals. *Metall. Mater. Trans. A* 31, 2219–2228.
- Reed, R.C., 2006. *The Superalloys Fundamentals and Applications*. Cambridge University Press, Cambridge.
- Sass, V., Feller-Kniepmeier, M., 1998. Orientation dependence of dislocation structures and deformation mechanisms in creep deformed CMSX-4 single crystals. *Mater. Sci. Eng. A* 245, 19–28.
- Sass, V., Glatzel, U., Feller-Kniepmeier, M., 1996. Anisotropic creep properties of the nickel-base superalloy CMSX-4. *Acta Mater.* 44, 1967–1977.
- Shi, D., Sui, T., Li, Z., Yang, X., 2022. An orientation-dependent creep life evaluation method for nickel-based single crystal superalloys. *Chin. J. Aeronaut.* 35, 238–249.
- Sun, F., Wang, S., Xie, Q., 2022. Role of particles and lattice rotation in tension–compression asymmetry of aluminium alloys. *Int. J. Plast.* 159, 103464.
- Tan, Y., Zang, C., Zhou, B., Wang, X., Petrov, E.P., 2018. Identification of crystal orientation for turbine blades with anisotropy materials. *Chin. J. Aeronaut.* 31, 410–418.
- Wang, J., Liang, J., Zhang, D., Peng, Y., Wen, Z., 2023. The effect of small orientation deviation from [001] to [011] on high-temperature creep properties of nickel-based single crystal. *Int. J. Plast.* 166, 103648.
- Wu, X., Dlouhy, A., Eggeler, Y.M., Spiecker, E., Kostka, A., Somsen, C., Eggeler, G., 2018. On the nucleation of planar faults during low temperature and high stress creep of single crystal Ni-base superalloys. *Acta Mater.* 144, 642–655.
- Xia, W., Zhao, X., Wang, J., Yue, Q., Cheng, Y., Kong, L., Zhang, Y., Gu, Y., Bei, H., Zhang, Z., 2023. New strategy to improve the overall performance of single-crystal superalloys by designing a bimodal γ' precipitation microstructure. *Acta Mater.* 257, 119200.
- Xiao, J., Jiang, W., Han, D., Li, K., Lu, Y., Lou, L., 2022. Evolution of crystallographic orientation and microstructure in the triangular adapter of grain continuator of a 3rd-generation single crystal superalloy casting during directional solidification. *J. Alloys Compd.* 898, 162782.
- Ye, Z., Li, C., Zheng, M., Zhang, X., Yang, X., Gu, J., 2022. In situ EBSD/DIC-based investigation of deformation and fracture mechanism in FCC- and L12-structured FeCoNiV high-entropy alloys. *Int. J. Plast.* 152, 103247.
- Yu, J., Li, J.R., Liu, S.Z., Han, M., 2020. Creep anisotropy in single-crystal superalloy DD6 near the [001] orientation. *Superalloys* 2020, 303–311.
- Yu, J., Li, J.R., Zhao, J.Q., Han, M., Shi, Z.X., Liu, S.Z., Yuan, H.L., 2013. Orientation dependence of creep properties and deformation mechanism in DD6 single crystal superalloy at 760°C and 785MPa. *Mater. Sci. Eng. A* 560, 47–53.
- Zhang, C., Wang, P., Deng, Y., Wang, X., Wen, Z., Lian, Y., He, P., 2024a. Influence of crystal orientations on the creep fracture of a nickel-based single crystal superalloy. *Int. J. Solids Struct.* 288, 112614.
- Zhang, J., Liu, Y., Cheng, Y., Wang, H., Sha, A., Duan, H., 2024b. Modeling of creep in nickel-based superalloy based on microtwinning mechanism. *Int. J. Plast.* 174, 103916.
- Zhang, Y., Liu, L., Huang, T., Li, Y., Jie, Z., Zhang, J., Yang, W., Fu, H., 2017. Investigation on remelting solution heat treatment for nickel-based single crystal superalloys. *Scripta Mater.* 136, 74–77.
- Zhou, H., Zhang, X., Wang, P., Lu, S., 2019. Crystal plasticity analysis of cylindrical holes and their effects on the deformation behavior of Ni-based single-crystal superalloys with different secondary orientations. *Int. J. Plast.* 119, 249–272.